Molded, Solid-State Biomolecular Assemblies with Programmable Electromechanical Properties

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(Received 25 March 2024; accepted 30 July 2024; published 27 September 2024)

Piezoelectric and ferroelectric technologies are currently dominated by perovskite-based ceramics, not only due to their impressive figures of merit, but due to their versatility in size and shape. This allows the dimensions of, for example, lead zirconium titanate and potassium sodium niobate, to be tailored to the needs of thousands of applications across the automotive, medical device, and consumer electronics industries. In this Letter, we significantly advance the performance and customization of biomolecular crystal (nontoxic, biocompatible amino acids, viz., *trans*-4-hydroxy-L-proline, L-alanine, hydrates of L-arginine and L-asparagine, and γ-glycine) assemblies by growing them as molded, substrate-free piezoelectric elements. This methodology allows for electromechanical properties to be embedded in these assemblies by fine-tuning the chemistry of the biomolecules and thus the functional properties of the single crystal space group. Here, we report the piezoelectric, mechanical, thermal, and structural properties of these amino acid-based polycrystalline actuators. This versatile, low-cost, low-temperature growth method opens up the path to phase in biomolecular piezoelectrics as high-performance, eco-friendly alternatives to ceramics.

DOI: 10.1103/PhysRevLett.133.137001

Introduction—Piezoelectricity presents as a linear electromechanical coupling in noncentrosymmetric structures that is exploited in areas as diverse as medical devices [1–3] and earthquake detection [4]. Their ability to harvest direct or ambient mechanical stimuli to convert to electrical energy means that they also have huge potential for self-powered wearable technologies [5,6] and eco-friendly electrical charging [7,8].

Biological materials [9], and molecular crystals [10], more broadly, tend to have inherent piezoelectricity due to their low symmetry, and crystallize in the solid state through different types of noncovalent intermolecular interactions, e.g., hydrogen bonds, halogen interaction, Van der Waals interactions, π stacking interactions, and ionic interactions [11]. Since the quantification of tangible piezoelectric responses in amino acid [12–16], peptide [17–21], and protein crystals [22,23], a shift is occurring toward modulating the electromechanical properties of biomolecular crystals using crystal engineering principles [24,25]. Alongside this, we have seen biomolecular crystals processed as stretchable films [26], hybrid nanogenerators [27,28], and composites [29], as well as efforts to orientate crystals and increase the piezoelectric response [29] using techniques such as cold sintering [24], nanoconfinement [30], and annealing [31].

What has been missing to date in the effort to accelerate the uptake of biomolecular crystals as high-performance commercial piezoelectrics, is a technique for growing polycrystalline assemblies in a repeatable and versatile manner at the macroscale. Prior to this Letter, biomolecular piezoelectric materials have been fabricated as single crystals [32], as thin films made via self-assembly [33] on conductive substrates, e.g., Cu [34], or on membrane- or polymer-based flexible materials [35]. To our knowledge this is the first time that a novel molded technique has been employed for the fabrication of amino acid-based piezoelectric materials as rigid molds. Amino acids are environmentally benign materials and decompose into nontoxic and biocompatible wastes making them inherently safer [36]. This technique allows us to have full control over the dimensions of the material enabling them to be easily incorporated into various electromechanical applications [2,37–41].

To date, freestanding flexible polyvinylidene fluoride (PVDF) nanocomposite films have been formed by drop casting the solution onto glass plate followed by thermal annealing and peeling off from the substrate [42]. Likewise, β-glycine and chitosan film have been fabricated by drop casting glycine: chitosan 0.8 : 1 in stochiometric ratios on a Petri dish and peeled off after growth was used as pressure sensor for bio applications [43]. It was recently reported that CBZ-Phe, a superhydrophobic self-standing material, was formed by drop casting thrice onto the polydimethylsiloxane substrate and later peeled off from it [44]. Here, we present a facile methodology for growing biomolecular crystals as stand-alone macroscopic piezoelectric elements without relying on substrates or polymers. While much of the literature has focused on polymer-centered flexible amino acid materials, these molded amino acid assemblies are the first to potentially challenge the wide variety of thick, rigid, and dimensionally diverse ceramic-based piezoelectric device components.

Our focus was to develop a standardized growth process of polycrystalline amino acids by a new molding technique, where solutions of polycrystalline amino acids were crystallized in silicone molds. In the case of conventional piezoelectric ceramic devices, the fabrication is time consuming and requires high temperature sintering and calcification processes, which are expensive and environmentally harmful [45]. Conventional piezoelectric lead-based materials [46] contain toxic lead-based components and upon degradation with time leach the toxic components into the environment. Further, fabrication of such materials is challenging and often expensive [47,48]. Alternative lead-free piezoelectric materials based on Ba [49], Bi [50–52], and Nb [53] also cause permanent environmental damage due to irreversible raw material extraction, material processing, and difficulty in recycling that results in a toxicological footprint [38,53–55]. The lower thermal stability and huge recycling costs of these lead-free competitors and their derivatives and the complicated processing of composite materials make them less sustainable as piezoelectric materials [44,56]. Moreover, alternative piezoelectric polymer materials like PVDF decompose into the extremely corrosive hydrogen fluoride by-product and have lower working temperature ranges, etc. [57–59].

Emerging biomolecular piezoelectric materials have been shown to outperform conventional piezoelectric materials in terms of eco-friendliness, cost-effectiveness, biodegradability, sensitivity, and mechanical strength [38,41,47,60–62]. In this Letter, we propose a new, easy-to-grow, tuneable, low-cost, and eco-friendly molding technique to make these biomolecular crystals suitable for employing in devices with minimal compromise in output performance.

Results—Table I shows the amino acids that were crystallized in this Letter [63], along with their space groups, CCDC identifiers, and the types of piezoelectric response that are permitted by their crystal symmetries.

TABLE I. List of amino acids, their space group, piezoelectricity permitted, and Cambridge Structural Database (CCDC) codes involved in the study.

Amino acid	Space group	Piezoelectricity permitted	CCDC code
γ-glycine	$P3_2$	Longitudinal, Shear	GLYCIN01
trans-4-hydroxy-L-proline	$P2_12_12_1$	Shear	HOPROL12
L-alanine	$P2_12_12_1$	Shear	LALNIN35
L-arginine dihydrate	$P2_12_12_1$	Shear	ARGIND11
L-asparagine monohydrate	$P2_12_12_1$	Shear	ASPARM

Two different space groups were chosen to see if naturally occurring longitudinal piezoelectricity would be amplified using this method, and also to see if orthorhombic structures would demonstrate a net longitudinal response. Orthorhombic single crystals can only demonstrate shear piezoelectricity, i.e., they will only generate charge with the application of a shear force, but polycrystalline films of orthorhombic amino acids have been shown to possess longitudinal piezoelectricity, likely due to the random orientation of crystallites [95].

Figure 1 shows these amino acids crystallized into macroscopic solid-state piezoelectric components. The

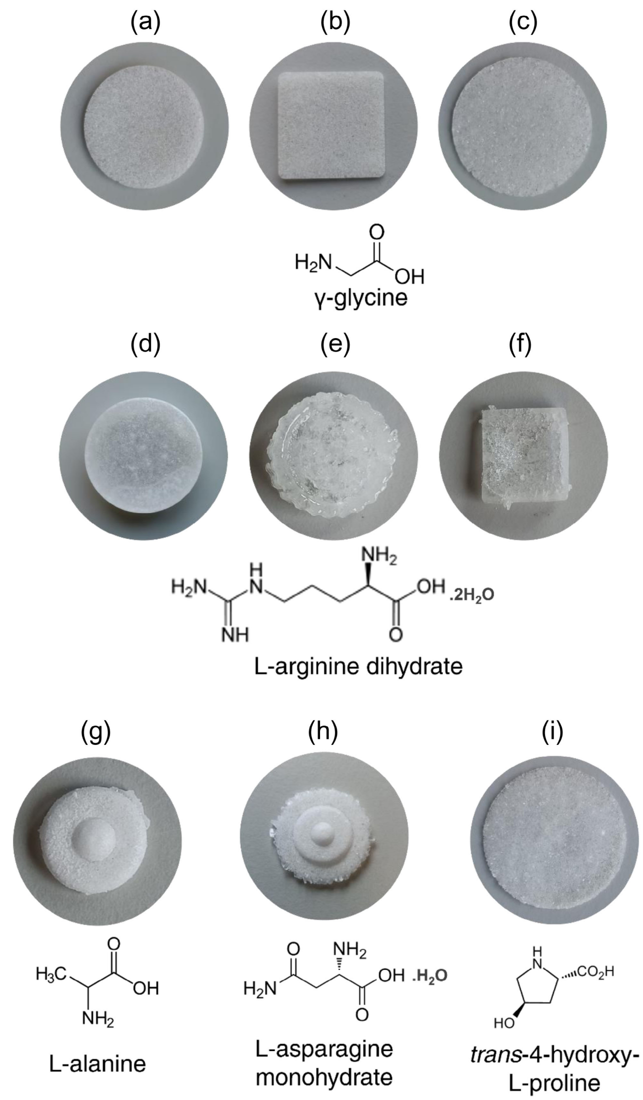

FIG. 1. A range of molded amino acid crystal assemblies and their chemical structures. (a) γ-glycine disc of diameter = 2.3 cm, (b) plate with sides 2.5 cm, (c) disc of diameter = 3.3 cm; (d),(e) L-arginine dihydrate disc of diameter = 2.3 cm; (f) plate with side 2.5 cm; (g) L-alanine disc of diameter = 1.5 cm; (h) L-asparagine monohydrate disc of diameter = 2.2 cm; (i) *trans*-4-hydroxy-L-proline disc of diameter = 3.3 cm.

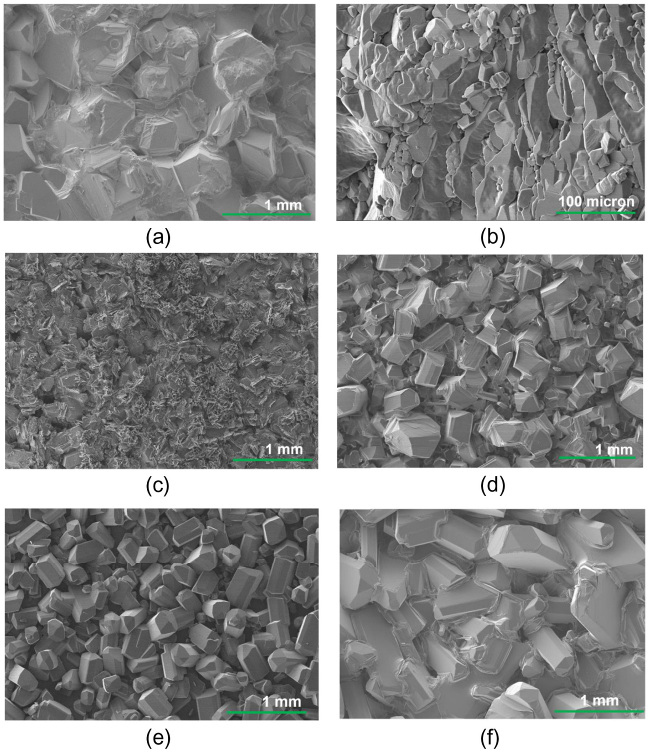

FIG. 2. SEM images revealing the morphology and orientation of the crystallites in the respective molded amino acid crystal assemblies (a) γ-glycine (1 mm), (b) (100 micron), (c) L-arginine dihydrate (1 mm), (d) L-alanine (1 mm), (e) L-asparagine monohydrate (1 mm), (f) *trans*-4-hydroxy-L-proline (1 mm).

polycrystalline structures form into the shape of the silicone mold, into which a supersaturated solution is poured and left to crystallize. In the main body of this Letter, we will focus on characterizing square plates and circular discs of amino acids, which are preferable for technological applications, but more complex structures are shown in Figs. 1(e), 1(g), and 1(h). The resulting polycrystalline molded assemblies have been optimized for various parameters, including solution concentration, volume, and time left to evaporate at room temperature. No by-products were formed after crystallization. The characterization of the formed structures was carried out using thermal analyses (Fig. S1) and powder x-ray diffraction (Fig. S2), followed by measurement of their longitudinal piezoelectric response.

At the macroscale, the structures are smooth along any dimension that has been in contact with the silicone during crystallization (outer edge and bottom surface), with the only roughness visible to the eye being on the upper surface that has been exposed to the air as the solvent evaporates. The SEM images (Fig. 2) show that the individual single crystals morph into each other at the microscale. As two single crystals form, they meet each other at an interface and continue to nucleate, forming a cohesive large-scale three-dimensional crystal assembly.

Figures 1(a) and 1(c) show circular discs and (b) shows plates of γ-glycine. γ-glycine crystals are known to have one of the highest reported Young's modulus of the proteinogenic amino acids [96], which results in thick stable discs and plates that are relatively easy to remove from the silicone casing. L-arginine dihydrate components are also easy to remove from molds [Figs. 1(d)–1(f)] but demonstrated a sticky gluelike appearance. L-alanine and L-asparagine monohydrate components are very fragile [Figs. 1(g) and 1(h)], which makes it difficult to carry out contact characterization. For these systems, modified molding techniques need to be employed to make them feasible for high-force applications. In contrast to this, crystals of *trans*-4-hydroxy-L-proline [Fig. 1(i)] form thinner plates that are more brittle than their counterparts, but when handled with care, all piezoelectric characterizations and measurements can be carried out. Figure 2 shows the microscale packing that contributes to these distinct physical properties. L-alanine [Fig. 2(d)] and L-asparagine monohydrate [Fig. 2(e)] grow more as assemblies of single crystals with sharp distinct edges that make the assemblies prone to breakage regardless of the mechanical properties of the crystals themselves. γ-glycine [Figs. 2(a) and 2(b)] and *trans*-4-hydroxy-L-proline [Fig. 2(f)] crystals fuse together during crystallization into a cohesive polycrystalline bulk that makes them the best candidates for further testing.

Figure 3 shows the average thicknesses and average longitudinal piezoelectric constant (d_{33}) of a series of γ-glycine and *trans*-4-hydroxy-L-proline structures. The cross-sample variation in mean piezoelectricity can be attributed to the natural self-assembly of the crystals that results in different crystal orientations in each sample. As piezoelectric measurements are taken as averages of the upright and inverted responses (Tables S1–S8), the standard deviations can also vary cross-sample due to varying degrees of sample roughness on the top surface. The responses in these sample sets are lower than the single crystal d_{33} response of γ-glycine [97,98], indicating a preferred alignment along the crystallographic a and b axes in these assemblies, which give rise to a piezoelectric response between 1 and 2 pC/N. However, they are higher than a number of monoclinic amino acids that were recently reported as aligned polycrystalline assemblies formed by mechanical annealing [31]. The thickness variation can also be attributed to sample roughness on the top surface. Further, the thickness of the molds can be varied by a layer-by-layer methodology where the previous layer can act as a seed for the new layer. No correlation was observed between thickness and piezoelectric response (Fig. S3), though we do not rule out a thickness dependence if roughness and multidirectional crystallite formation can be eliminated in the future. For γ-glycine discs of diameter = 2.3 cm, the sample thickness varies between 1.44 and 1.79 mm, with d_{33} values varying between 0.51 pC/N and 1.12 pC/N and similarly for γ-glycine discs of diameter = 3.3 cm, the range of thicknesses is 1.83–2.28 mm and d_{33} is in the range of 0.41–2.09 pC/N. γ-glycine plates demonstrate thicknesses of 1.79 to

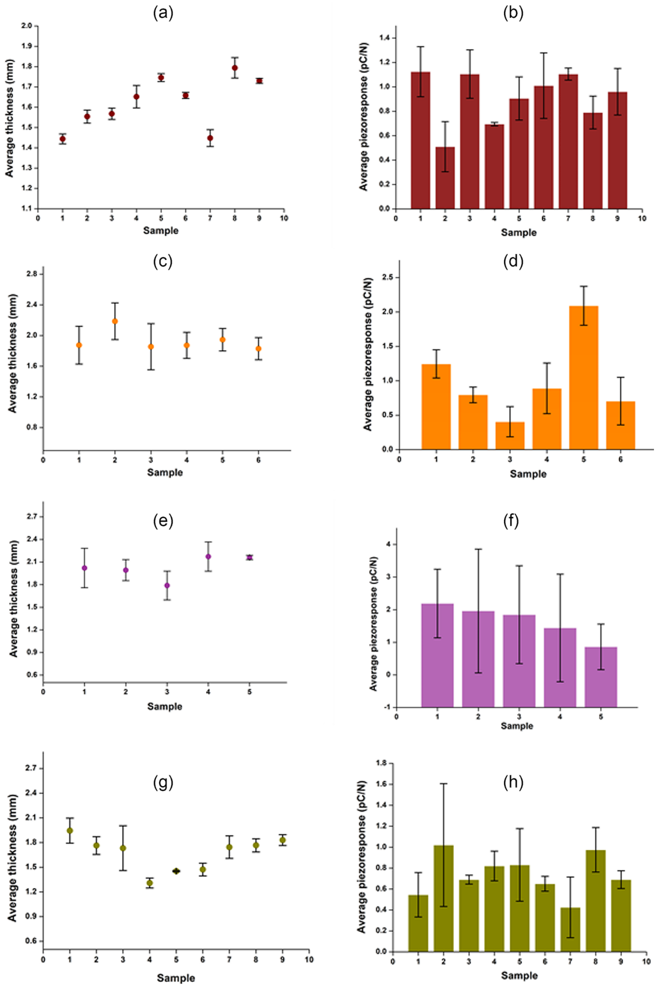

FIG. 3. Average thickness (in mm; left) and average piezoelectric measurements (in pC/N; right) of (a),(b) γ-glycine (disc of diameter 2.3 cm); (c),(d) γ-glycine (disc of diameter 3.3 cm); (e),(f) γ-glycine (plate of sides 2.5 cm); (g),(h) *trans*-4-hydroxy-L-proline (disc of diameter 3.3 cm).

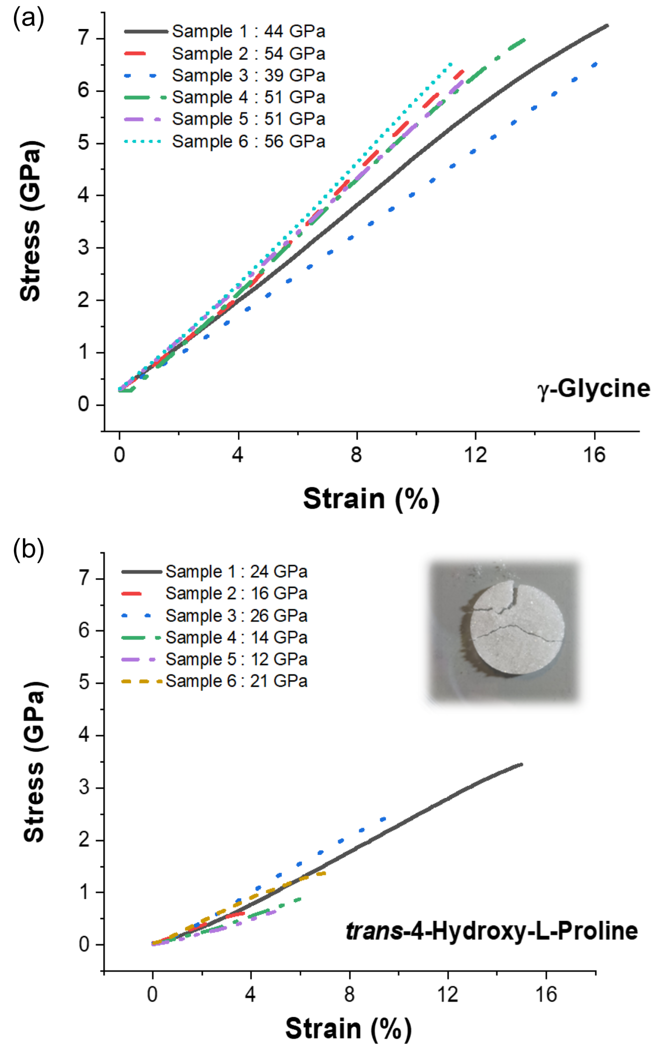

FIG. 4. Stress-strain curves for polycrystalline discs of (a) γ-glycine, (b) *trans*-4-hydroxy-L-proline (inset shows the disc after compression testing).

2.17 mm and exhibit piezoelectric responses from 0.86 to 2.19 pC/N. *trans*-4-hydroxy-L-proline discs of diameter = 3.3 cm show thicknesses in the range of 1.31–1.94 mm within an average piezoresponse range of 0.43–1.02 pC/N. For γ-glycine, a maximum average piezoresponse of 11.11 pC/N for discs of 2.3 cm diameter and 10.3 pC/N for discs of 3.3 cm diameter was recorded. *trans*-4-hydroxy-L-proline discs demonstrate a maximum average piezoresponse of 2.2 pC/N for a disc of 3.3 cm diameter. This demonstrates that, with favorable directional alignment, these polycrystalline assemblies of γ-glycine can match and even exceed their single crystal longitudinal response. For *trans*-4-hydroxy-L-proline this maximum response exceeds our previous reports on these crystals as grown on flexible copper substrates [95]. It also confirms that orthorhombic biomolecular crystals should not be eliminated as candidates for longitudinal sensing and actuation, as polycrystalline fabrication allows us to exploit their single crystal shear piezoelectricity.

Figure 4 shows the linear regions of the experimental stress-strain curves for γ-glycine and *trans*-4-hydroxy-L-proline assemblies measured using compression testing. When the entirety of the data from each sample was plotted, it showed multiple sharp spikes, which corresponded to breaks in the sample. The polycrystalline nature of the samples means that multiple cracks can form in the material before a final absolute fracture point is reached. The initial linear region until the first crack was taken and plotted. The slope of these lines tells us Young's modulus for each sample, or for anisotropic crystals, the elastic stiffness in the direction of the applied force [99].

The two amino acids show distinct average Young's moduli of 49 and 19 GPa for γ-glycine and *trans*-4-hydroxy-L-proline, respectively, which is consistent with both previous studies on their supramolecular packing and mechanical behavior, and their handleability in the lab for this Letter. There are observable variations in each sample's

TABLE II. Average capacitance (in pF) and piezoelectric voltage constants (in mV m/N) of molded amino acid assemblies.

Sample	Average capacitance (pF)	g_{33} (mV m/N)
γ-glycine (plate; side of 2.5 cm)	0.69	29
γ-glycine (disc; diameter of 2.3 cm)	0.51	28
trans-4-hydroxy-L-proline (disc; diameter of 3.3 cm)	0.4	26
γ-glycine (disc; diameter of 3.3 cm)	1.77	7

deformation profile—both in terms of the slopes of the elastic region and the maximum strain before fracture. At the macroscale, the sample was consistently placed in the same location so that the presser would make contact with the center of the sample. However, at the microscale, it is much more difficult to ensure consistency in location, as the grain boundaries cannot be seen visually. Thus, cracks could propagate sooner or later, depending on if pressure was applied to a "weak point" in the assembly.

The average capacitance and piezoelectric voltage constants of the same sample forms as above are shown in Table II. The γ-glycine discs and plates have an average capacitance (C) of 0.510 pF (for 2.3 cm diameter), 1.77 pF (for 3.3 cm diameter), and 0.696 pF (sides 2.5 cm). The average capacitance (C) of *trans*-4-hydroxy-L-proline discs was found to be 0.418 pF (for 3.3 cm diameter). This low capacitance is typical of biomolecular crystals and has been shown to endow high sensitivity for piezoelectric vibration detection [38]. From the capacitance, thickness, and piezoelectric response of our samples (Table S9), we can calculate their longitudinal piezoelectric voltage constant, g_{33}, in mV m/N (Table I), using the below equation,

$$g_{33} = \frac{d_{33}}{\varepsilon\varepsilon_0}.$$

The voltage constants confirm that this methodology would be especially effective for batch γ-glycine (disc; diameter of 2.3 cm) processing of eco-friendly energy harvesters. The voltage constants range from 7 to 29 mV m/N for these samples with d_{33} values of 0.5 to 2 pC/N, compared to 40 mV m/N for lead zirconium titanate ceramics, and 240 mV m/N for PVDF polymers. For the highest-recorded d_{33} responses, we estimate the voltage constants of 298 mV m/N and 131 mV m/N for γ-glycine discs with diameters of 2.3 and 3.3 cm, respectively, and 91 mV m/N for *trans*-4-hydroxy-L-proline (disc; diameter of 3.3 cm) (Table S9).

Figure 5 shows the open circuit voltage obtained for each molded piezoelectric element measured by applying tapping forces on the samples. The electroded molded amino acids were subjected to manual tapping to replicate real-world mechanical stimuli, which results in different voltage peak shape and magnitude. A subjective range of tapping intensities were imparted onto the structures, connected to digital oscilloscope, to understand the sensitivity and voltage characteristics. The assemblies were very sensitive to even gentle manual tapping and provided voltage in the range of a few millivolts. For γ-glycine and *trans*-4-hydroxy-L-proline molds, we obtained a maximum peak-to-peak voltage of 1120 and 700 mV, respectively (Fig. S4).

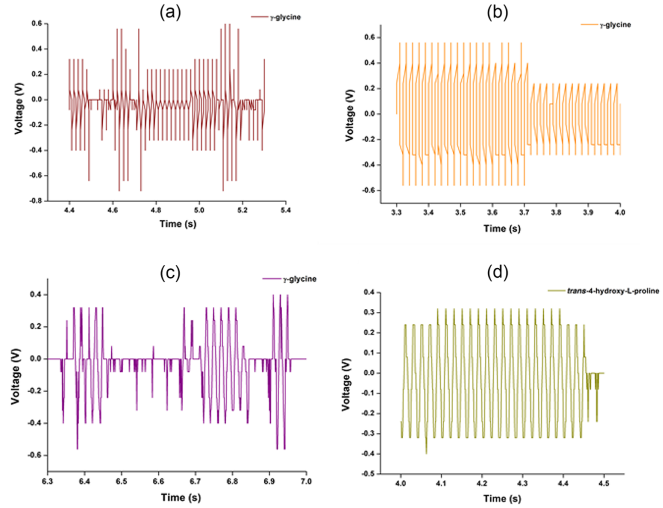

FIG. 5. Open circuit voltage of molded amino acid crystal assemblies. (a) γ-glycine (disc of diameter 2.3 cm), (b) γ-glycine (disc of diameter 3.3 cm), (c) γ-glycine (plate of sides 2.5 cm), (d) *trans*-4-hydroxy-L-proline (disc of diameter 3.3 cm).

Looking toward applications of these materials, it is evident from the respective differential scanning calorimetry (DSC) endotherms (Fig. S1) that γ-glycine piezoelectric elements will be operational between a temperature range of 298–448 K, as γ-glycine undergoes a phase transformation whose onset occurs around 458 K. Similarly, molded *trans*-4-hydroxy-L-proline crystals should be operational between a wider temperature range 298–568 K above which the decomposition of *trans*-4-hydroxy-L-proline is initiated (Fig. S1).

Conclusion—We have successfully grown polycrystalline amino acid assemblies in a facile, scalable, substrate-free manner. This allows these functional materials to be fabricated into any desired shape and size for sensing, actuation, and energy harvesting applications. We have shown that amino acid polycrystalline assemblies can be grown using an evaporative method onto custom silicone molds so that the resultant bulk assembly can act as a standalone component. Compared to ceramic device fabrication [45], our methodology contains only a few steps: (1) weighing the starting materials, (2) mixing to form a saturated homogeneous solution, (3) depositing into the mold and subsequently slowly evaporating at room temperature, and

(4) demolding and air drying. Moreover, these eco-friendly piezoelectric elements demonstrate a maximum peak-peak voltage of 1.1 V and can operate at high temperatures.

Our reported longitudinal responses, ranging from 0.02–3.09 pC/N, are similar to those of polycrystalline lysozyme films [100], which also demonstrated structurally induced longitudinal piezoelectricity of 0.67–1.34 pC/N. Our high-concentration molding technique could also be used on stable crystals of large biomolecules such as lysozyme that demonstrate technologically significant piezoelectricity [22] but suffer from technological drawbacks such as random orientation or low density of crystallites, large standard deviations between samples, and electroding issues due to surface roughness [63,100].

Our future goal is to work with the more fragile amino acids shown in this Letter to enhance their macroscopic mechanical strength [63]. This versatile molding technique can be extended to other eco-friendly molecular crystals as we move toward phasing out environmentally damaging ceramic piezoelectrics.

Acknowledgments—S. B., K. H., and S. G. would like to acknowledge funding from Science Foundation Ireland under grant No. 21/PATH-S/9737. S. G. acknowledges Science Foundation Ireland Award No. 12/RC/2275_P2. S. G. and T. R. are funded by the European Union under ERC Starting Grant No. 101039636. The authors thank Dr. Eoghan Cunnane and Clíona McCarthy for the use of their compression testing facilities.

Data availability—The data that support the findings of this study are available in the Supplemental Material [63].

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